

RESULTS

Results

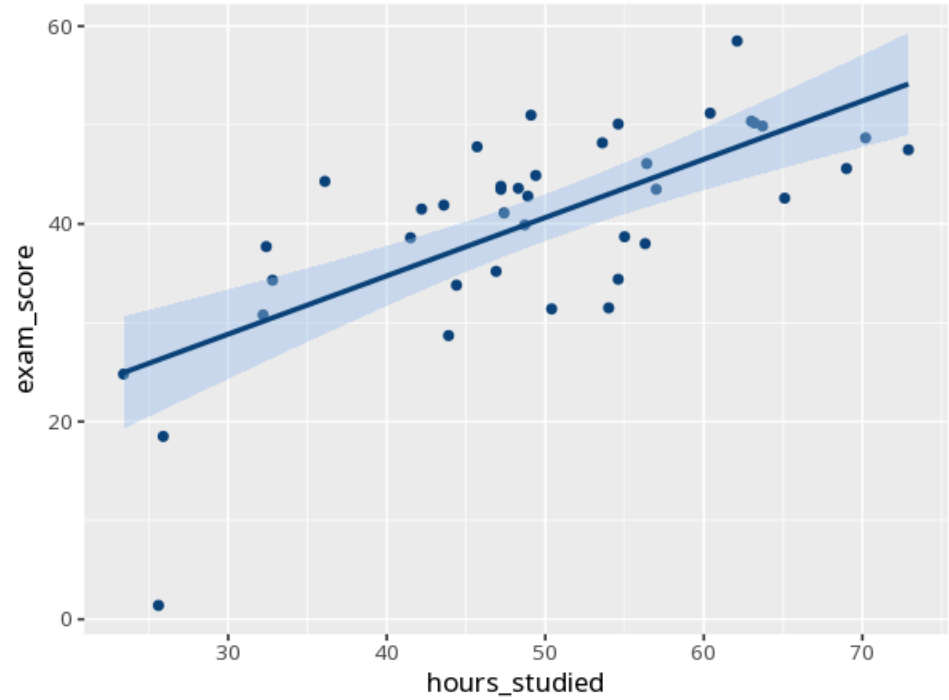
01 · PEARSON CORRELATION

linear association of two numeric variables

Table. Pearson correlation

Pair	r	95% CI	t	df	p	N
hours_studied – exam_score	0.70	[0.50, 0.83]	6.07	38	<.001	40

Figure. Relationship



Understanding the numbers

Pearson's r. The strength and direction of the linear relationship, from -1 to +1. Here, $r(38) = .70$, 95% CI [.50, .83] - a large positive relationship.

95% CI. The precision of the r estimate: the range of correlations plausible at the chosen confidence level. Here, the 95% CI for r is [.50, .83].

t. The test statistic testing whether r differs from zero. Here, $t(38) = 6.07$.

df. The degrees of freedom for the correlation test ($N - 2$). Here, $df = 38$.

p. The probability of a correlation this far from zero (or further) if the two variables were truly unrelated. Here, $p < .001$.

N. The number of complete pairs (cases with both variables present) used. Here, $N = 40$ complete pairs.

How to read this test

r ranges from -1 to $+1$: the sign is the direction, the magnitude the strength ($\sim .1$ small, $.3$ medium, $.5$ large). p tests whether it differs from zero; the 95% CI shows the precision. r measures only linear association — a strong curved relationship can give r near 0, and r is sensitive to outliers, so always inspect the scatterplot (use Spearman's r_s for monotonic-but-nonlinear or outlier-affected data). Correlation does not imply causation. Your significance threshold (α) is 0.05.

APA template: hours_studied and exam_score were correlated, $r(38) = .70$, $p < .001$, 95% CI [0.50, 0.83].

Statistical basis: Pearson product-moment correlation coefficient (r) (Pearson, K. (1895). "Note on regression and inheritance in the case of two parents." Proceedings of the Royal Society of London, 58, 240-242.); r effect-size benchmarks ($.1/.3/.5$) (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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